



LQG solution for POMDP without estimating states

A Minimum Variance Approach

Ranjan Sarkar, Prabhat K. Mishra

Department of Artificial Intelligence, Indian Institute of Technology Kharagpur



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Abstract

We compute constrained stabilizing control directly from the noisy measurements and previously applied controls, without explicitly estimating the hidden states.

Background

Linear POMDP (Partially Observable Markov Decision Process)

$$x_{t+1} = Ax_t + Bu_t + \xi_t, \quad y_t = Cx_t + \omega_t. \quad (1)$$

- The initial state follows the prior distribution, $x_0 \sim \mathbb{P}_0(\hat{x}_0^-, \Sigma_0^-)$
- The process and measurement noises satisfies $\xi_t \sim \mathcal{N}(0, Q), \quad \omega_t \sim \mathcal{N}(0, R).$
- With perfectly measured states, the LQG solution is $u_t = Kx_t$

where K is the **state feedback gain** obtained from the discrete algebraic Riccati equation (DARE).

Optimizing MSE cost (Stochastic)

$$\begin{aligned} & \underset{\hat{u}_t}{\text{minimize}} \quad \mathbb{E} [\|u_t - \hat{u}_t\|^2 \mid y_{0:t}, u_{0:t-1}] \\ & \text{subject to} \quad \text{POMDP (1)} \\ & \quad \quad \quad u_t = Kx_t \end{aligned}$$

Minimum-variance duality [1, 2] converts this stochastic cost into a deterministic LQR cost over dual variables.

Direct LQG Control Estimation

Dual Dynamics

$$z_{i+1} = A^\top z_i + C^\top \alpha_{i+1}, \quad z_0 = K^\top + C^\top \alpha_0 \quad (2)$$

Dual LQR Optimization (Deterministic)

$$\begin{aligned} & \underset{\alpha_{0:t}}{\text{minimize}} \quad \text{tr} \left[z_t^\top \Sigma_0^- z_t + \alpha_t^\top R \alpha_t + \sum_{i=0}^{t-1} (z_i^\top Q z_i + \alpha_i^\top R \alpha_i) \right] \\ & \text{subject to} \quad \text{dual dynamics (2)} \end{aligned}$$

Optimal Control

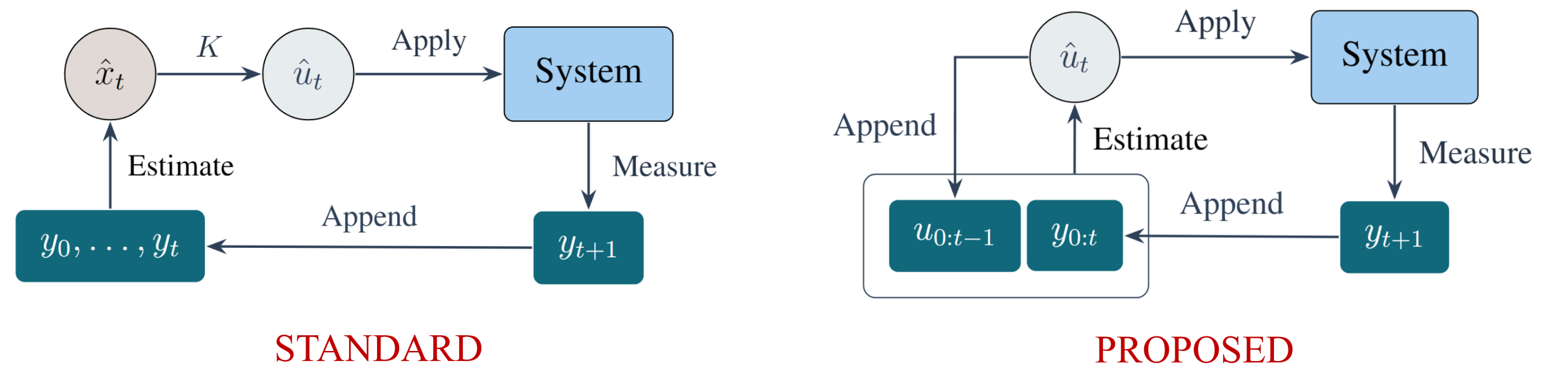
$$\hat{u}_t = z_t^\top \hat{x}_0^- + \sum_{i=0}^{t-1} z_i^\top B u_{t-i-1} - \sum_{i=0}^t \alpha_i^\top y_{t-i}$$

The optimization problem can be solved directly using **CasADi** package in python.

Future Direction

- Current framework uses all historical data. A moving horizon formulation can enable scalable implementation for long time horizons.
- Establish theoretical guarantees for the convergence of the control estimation error variance.

Architectural Comparison

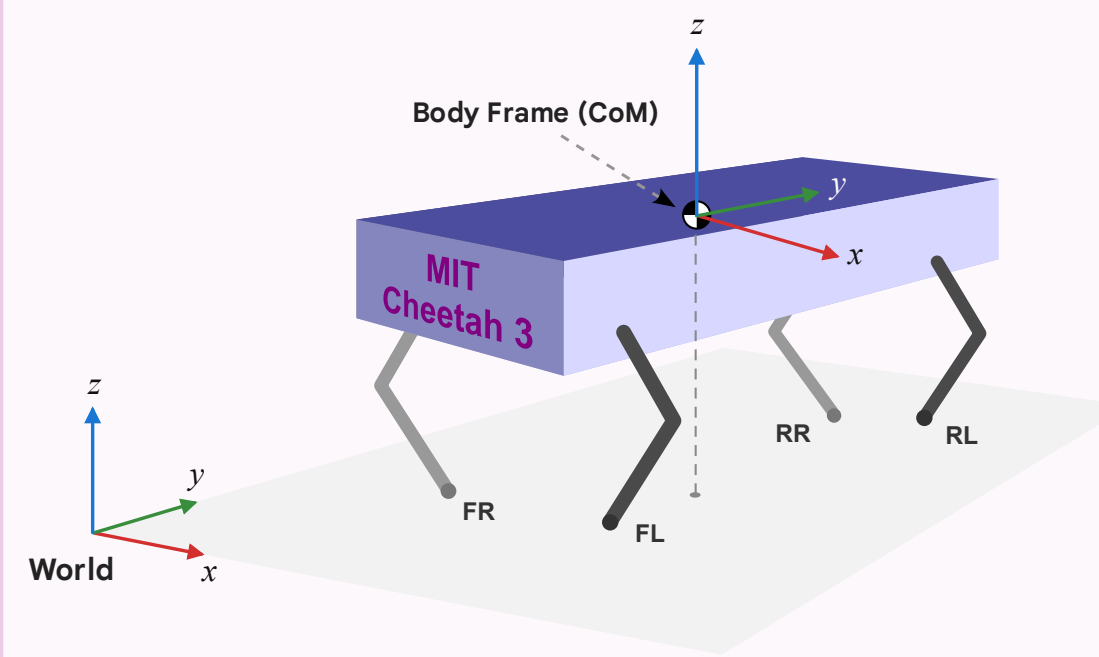


Problem with Standard Approach

Standard approach relies on:

- Separation Principle** (State estimation $\rightarrow$ LQR control)
- It breaks down when systems deviate from unconstrained, linear models, except in a few special cases.

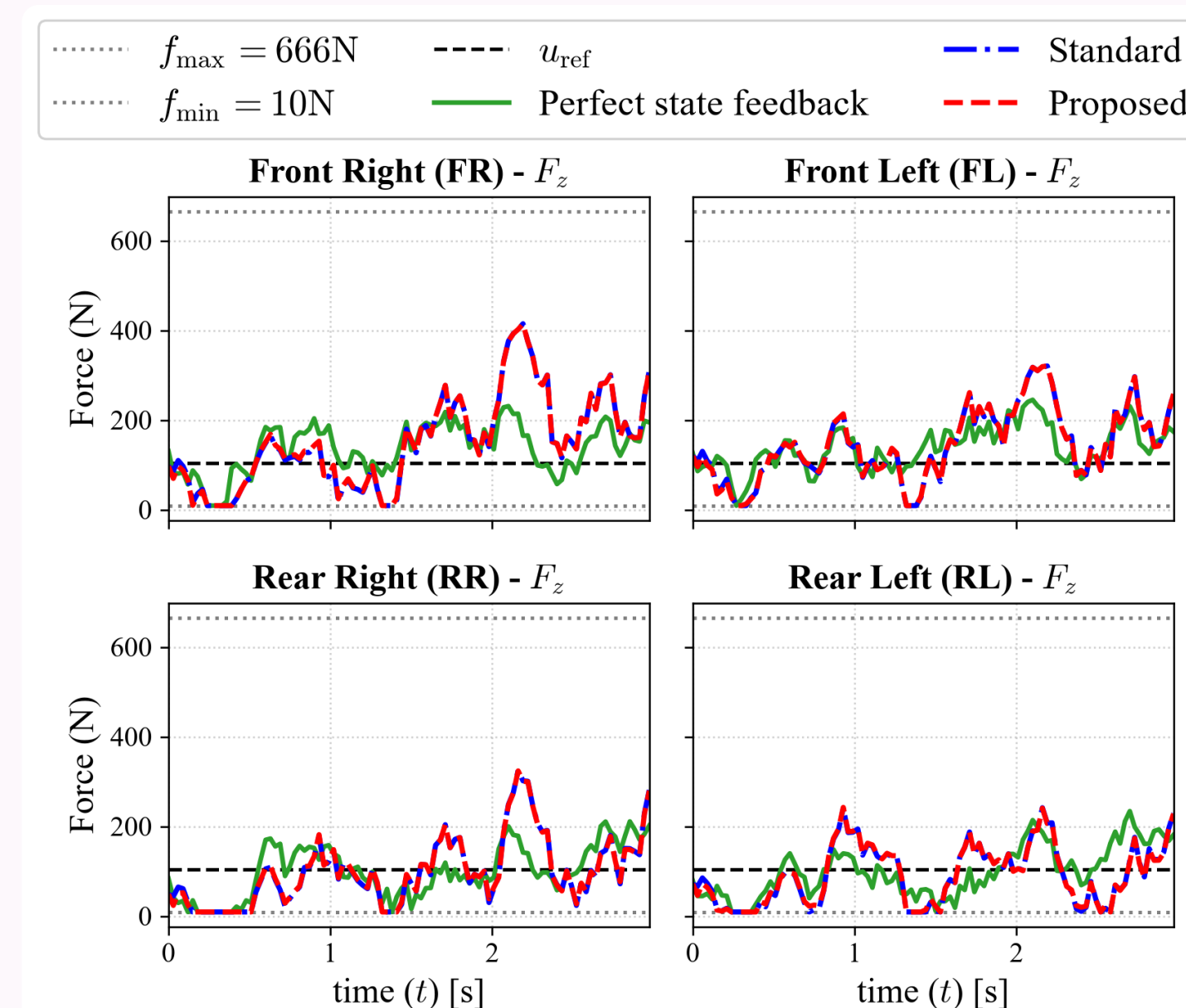
Experiment: Reference tracking by MIT Cheetah 3 [3]



- 13 state simplified rigid-body model.
- 12 ground reaction force inputs.
- 6 noisy measurements (first 6 states).
- Standing reference height: 0.45 m.
- Sample rate $\Delta t = 0.03$ second (s).
- 100 timesteps = 3 s.

- The **standard approach** first computes an unconstrained LQR control from the estimated state and then applies saturation function.
- The **proposed controller** can enforce force constraints inside the dual optimization problem.

Numerical Results (Initial)

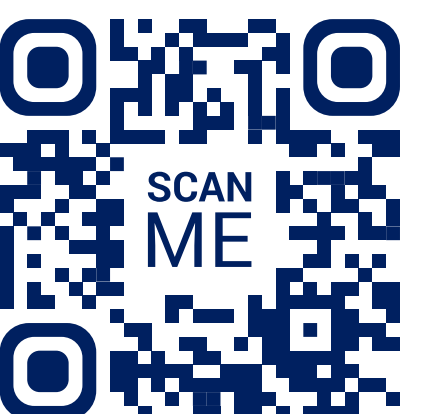


Insights

- The plots illustrate the vertical ground reaction forces (F_z) across all four legs (FR, FL, RR, RL) over 100 timesteps.
- The standard approach and the proposed approach are giving almost same control trajectories in this experiment.
- Due to stochastic disturbances, the computed control try to stabilize but oscillates toward the steady-state reference control (u_{ref}).

References

- [1] Åström, "Introduction to Stochastic Control Theory", 1970.
- [2] Mishra et al., "Minimum variance constrained estimator", *Automatica*, 2022.
- [3] Carlo et al., "Dynamic Locomotion in the MIT Cheetah 3 through Convex Model-Predictive Control", *IROS*, 2018.



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